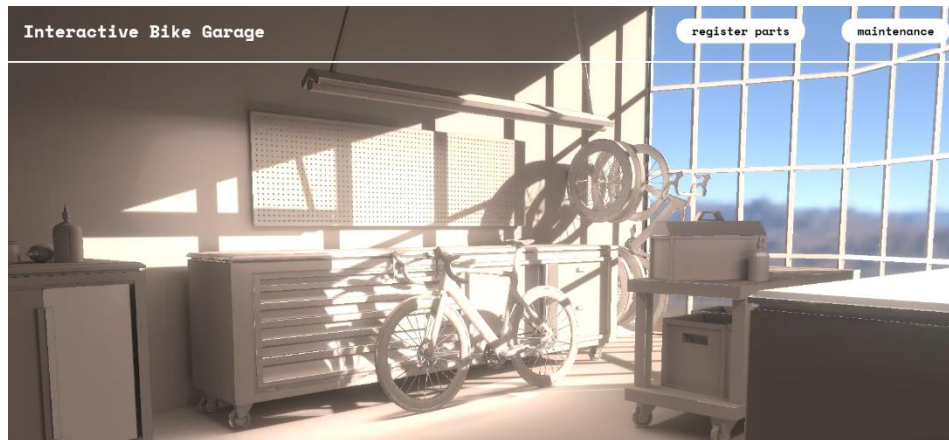

URL of the project (desktop and mobile):

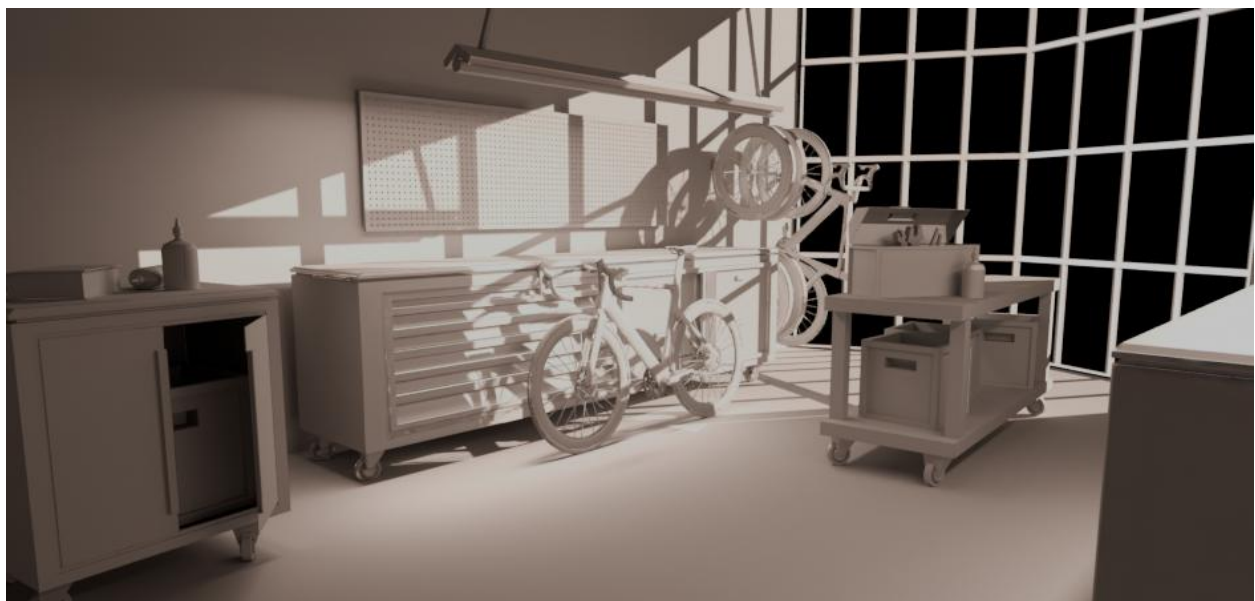
https://emilebedard.github.io/cart263/Projects/project2/interactive_bike_garage_ii/

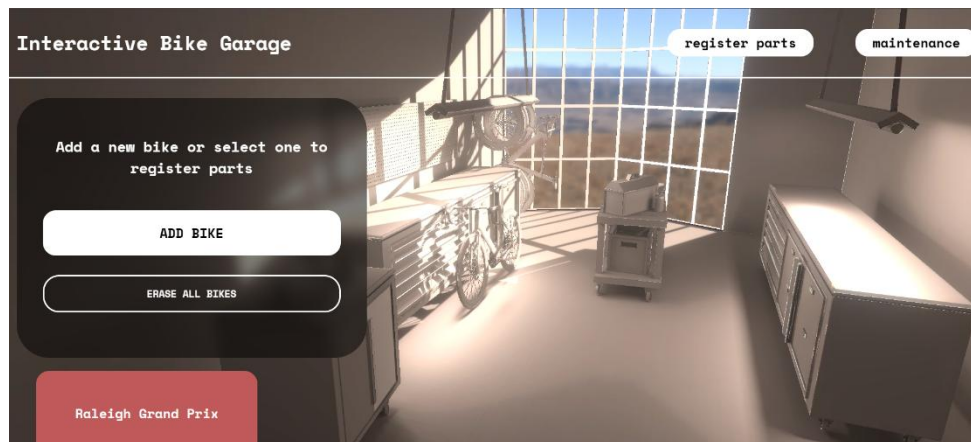


Main landing page

The main atmosphere of the web experience was the whole idea of the second project: offer an immersive 3D modern scene of a bicycle workshop. I modeled everything except for the bike and decided on the layout and composition of the scene based on my own preferences. I would like to imagine this space as my own future workshop.

A big inspiration throughout my studies for me was the “atelier rouge” by French painter Matisse. This minimalist painting shows the artist’s space and the singularity of it always inspired me to see my own space for creation in a singular way too, waiting for project. I wanted to achieve a similar “dream” creation space with Interactive Bike Garage II.





Layout and HTML hierarchy

My next challenge was to keep as much as possible from my previous hmtl/css layout and maximize my time and energy on the 3D with Three.js. I still feel not completely sure about the styling but integrating the 3D canvas seamlessly was not easy. For the next version, I will start from scratch and build on top of the canvas to really make a new styling and UI interface that feels much more integrated and cohesive.

```
function addAndRun(loadedObjArray) {
  console.log(loadedObjArray);
  //garage
  const garageScene = loadedObjArray[0].scene; //load blender data and place in the scene
  scene.add(garageScene);

  //cabinet
  const cabinetObj = loadedObjArray[1].scene; //load blender data and place in the scene
  scene.add(cabinetObj);

  //workbench
  const workbenchObj = loadedObjArray[2].scene; //load blender data and place in the scene
  scene.add(workbenchObj);

  //camera from blender---
  let blenderCamera = garageScene.children.find(child => child.isCamera);

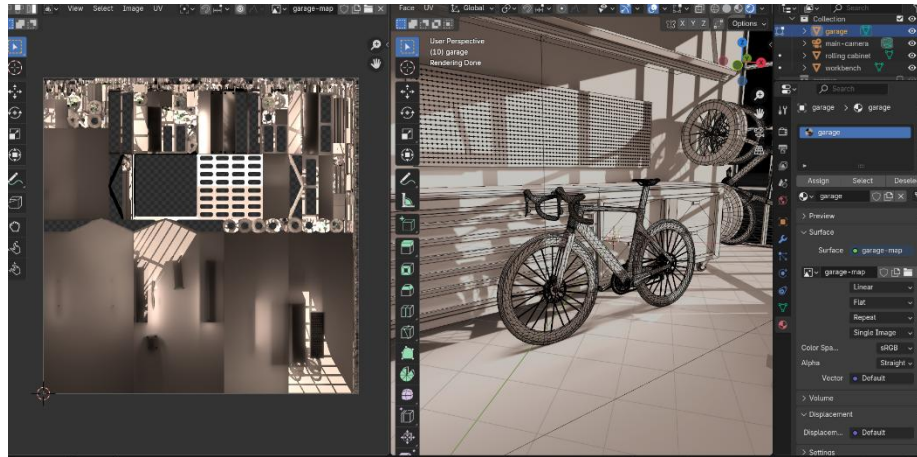
  //change between initial and blender cam here
  let usedCam = blenderCamera;
  if (usedCam === blenderCamera) {
    blenderCamera.name = "blenderCamera";
    camera = blenderCamera;

    const isRegistryPage = window.location.pathname.includes('parts_registry.html');
    if (isRegistryPage) {
      const registryPos = new THREE.Vector3(-17, 15, 40);
      camera.position.copy(registryPos);
      originalBlenderPos.copy(registryPos);
    } else {
      originalBlenderPos.copy(blenderCamera.position);
    }

    // Update the aspect ratio to match the current html container dimensions stated in the setup container function
    camera.aspect = sizes.width / sizes.height;
    camera.updateProjectionMatrix();
    renderPass.camera = camera;
  }
}
```

Importing and integrating GLTFs in THREE.JS

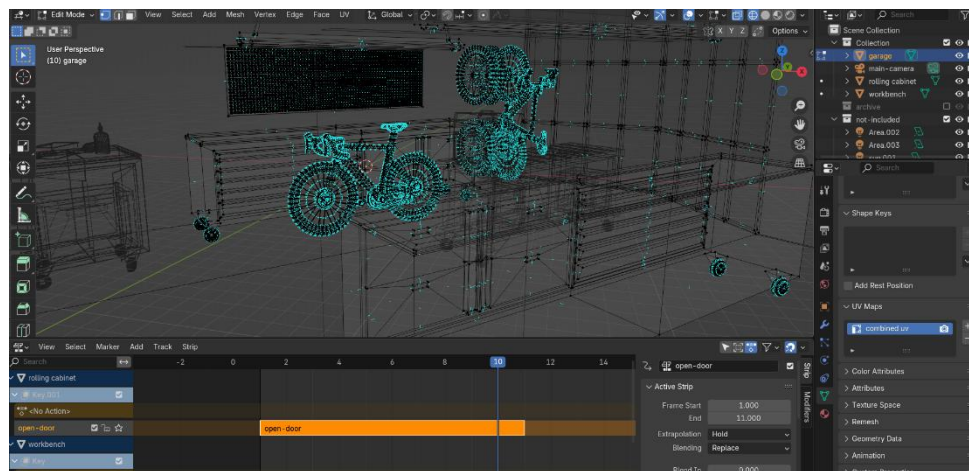
Importing the GLTF format and retrieving data, animation, scenes, lights and cameras was very tricky. It was imperative to keep a good file system and to really organize my objects name to be able to point them out easily afterwards. The camera logic was especially delicate because I needed to redirect it from the Three.js interface to store it in a variable, rename it and then attach to my renderer not only once it was loaded from my try/catch asynchronous function.



Lightmap baking and UV unpacking

Lightmap baking and UV unpacking were familiar concept for me with the previous experience that I had with Blender, but I never fully implemented them in a project before this one. It was tricky to figure out the technical manipulations and procedures but once these constraints were out the way, I really had a lot of fun to play with the light and achieve a much more realistic look for my scene.

The look that I went for is heavily inspired by 3D artist Jo-Anie Charland, Baeige, that combines light color palettes and architectural scenes to evoke serenity in those digital liminal spaces. This calm and dehumanized architectural space was exactly what I had in mind for this project.



Normals and face orientation

Finally, normals and 3D face orientations were also a significant speed bump in my Blender 3D Pipeline. As I explain in the video, importing external 3D models and assets can always be tricky and this bicycle 3D model was no exception. I had to rework all the faces orientation manually to achieve realistic light reflections with the rendering and global illumination.